

Teaching the Transportation Problem with Spatial Data

An Active Learning Exercise for Operations Research & Supply Chain Management

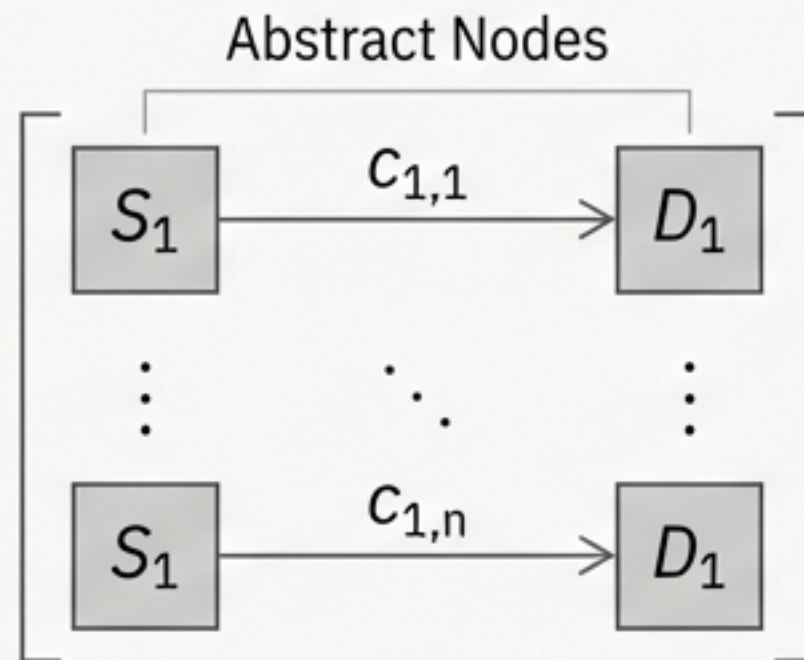
Synthesised from the work of Jyotirmoy Dalal (Indian Institute of Management Lucknow), published in *INFORMS Transactions on Education* (2022).



Bridging the gap between abstract textbook matrices and the messy reality of last-mile logistics.

The Pedagogical Gap: Why Abstract Matrices Fail to Engage

The Textbook Approach



- Standard Operations Research (OR) curricula rely on pre-given matrices.
- Focuses heavily on theoretical properties (integrality, network simplex).
- Ignores the complexity of data generation.
- Result: Disconnect between 'solving the model' and 'designing the network'.

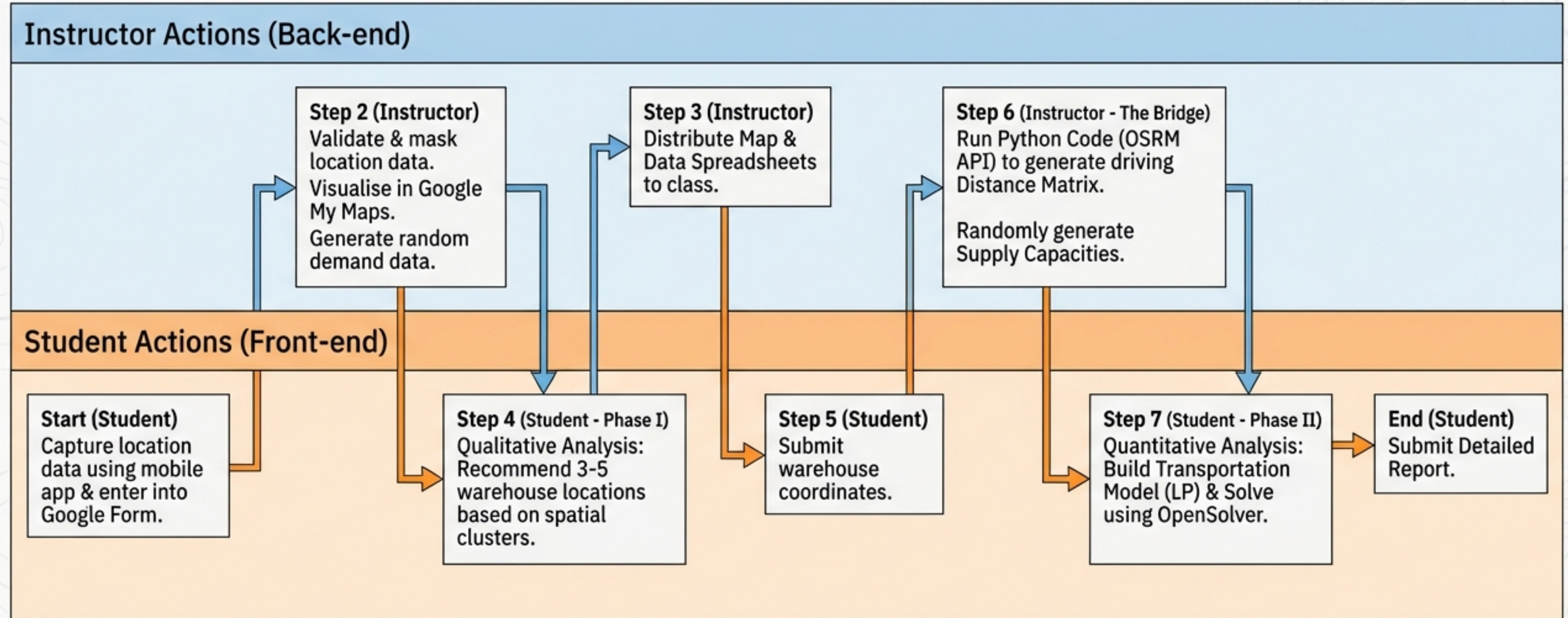
The Real-World Gap



- Post-COVID-19 logistics: Phenomenal growth in online market demand.
- Student Demand: Business students crave practical applicability over methodological nuances.
- The Solution: Students become the nodes. By utilising their own geographic locations, the class generates a realistic spatial dataset, fostering immediate emotional investment.

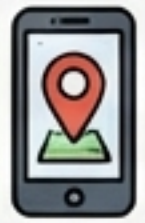
The Exercise Architecture: A Two-Phase Playbook

Timeline: Approx. 2 Weeks | Structure: Student-Instructor Handshake



The Toolkit: Open Source & Accessible

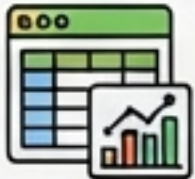
Student Requirements (Front-End Tools)



Smartphones: GPS Apps to capture Latitude/Longitude.

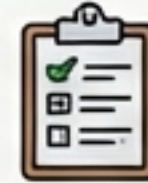


Google My Maps: For visualising demand clusters and spatial planning.



Spreadsheet Software: Excel or LibreOffice Calc.

Instructor Requirements (Back-End Tools)



Google Forms: For efficient data aggregation.



Python Script: *calcDM.py* (Provided) to automate distance calculations.



OSRM API: Open Source Routing Machine for real road network data.



OpenSolver: Excel add-in to bypass the standard 200-variable limit.



Mise en place

Step 1: Crowdsourcing the Network

Converting students into data points.

The Task: Students capture their current Latitude/Longitude using a GPS app and submit it via a Google Form.

Rich Data: Beyond location, students rank preferences for four product categories:

1. Electronics
2. Books
3. Apparel
4. Grocery

Why it Works: In a remote or hybrid class, students are dispersed. This creates a realistic, pan-national distribution of “customers”.

Field	Value
Enter Name (optional)	XXXXX
Enter State Name	Andhra Pradesh
Enter Pincode (digits only)	517502
Enter Latitude of your current location (up to 8 decimal point precision)	13.619065
Enter Longitude of your current location (up to 8 decimal point precision)	79.324446

Self-Generated Spatial Data

Rank	Electronics	Book	Apparel	Grocery
First Choice	X			
Second Choice			X	
Third Choice		X		
Fourth Choice				X

Data Prep: Validation, Masking, and Demand Logic

Instructor Actions (Back-end)



Privacy & Validation

Masking: Real coordinates are perturbed by adding a random fraction to entries.

Validation: Import raw CSV to Google My Maps to spot errors (e.g., missing decimals).



The Logic Engine

Converting text preferences to numeric demand.

```
IF(Choice="Books", 4,  
IF(Choice="Electronics", 3...))  
RANDBETWEEN(350, 450) if Rank=1
```

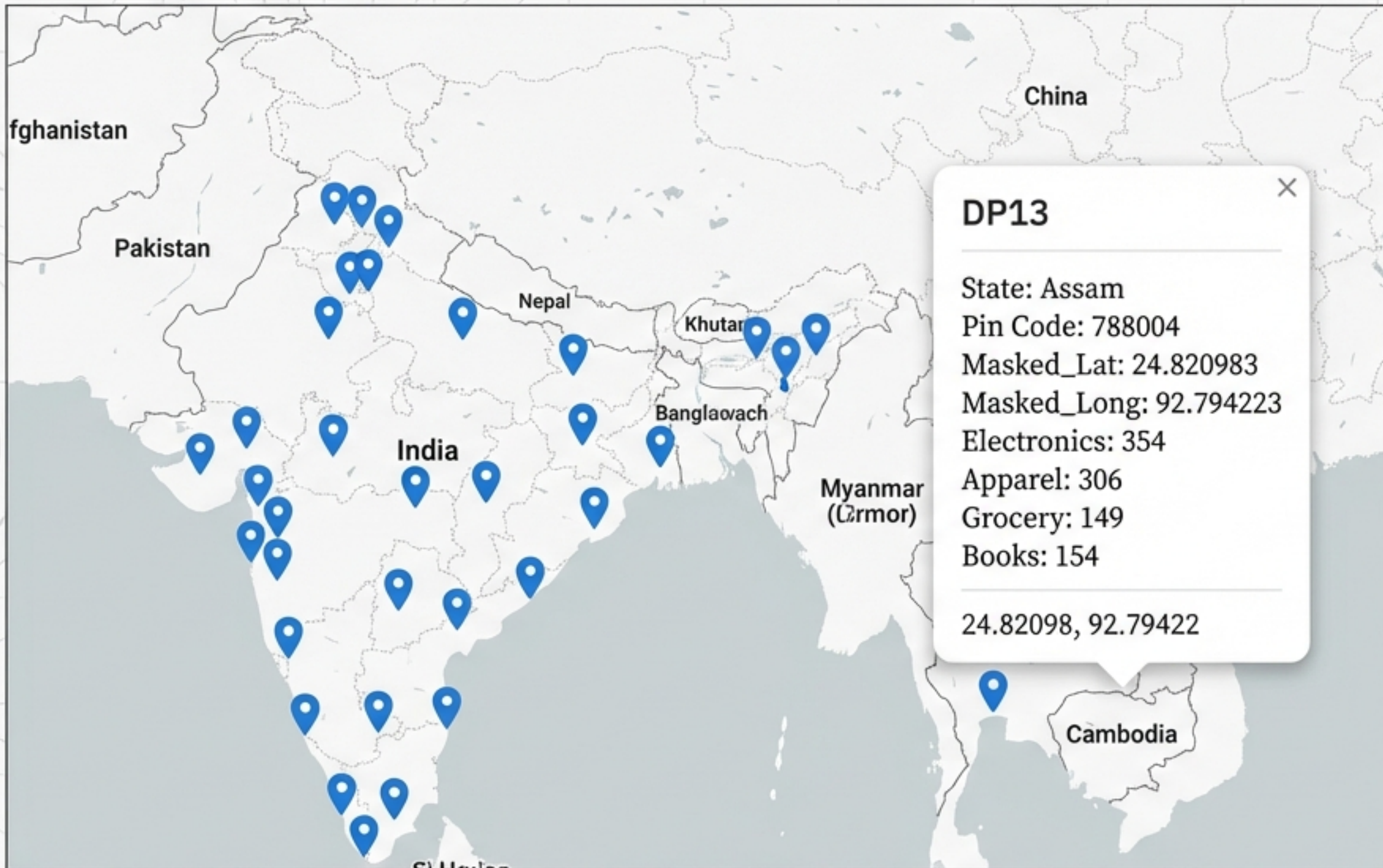


The Output

A clean dataset with 100+ Demand Points (DP1...DP101) ready for distribution.

Mise en place

Phase I: Visualising the Demand Landscape

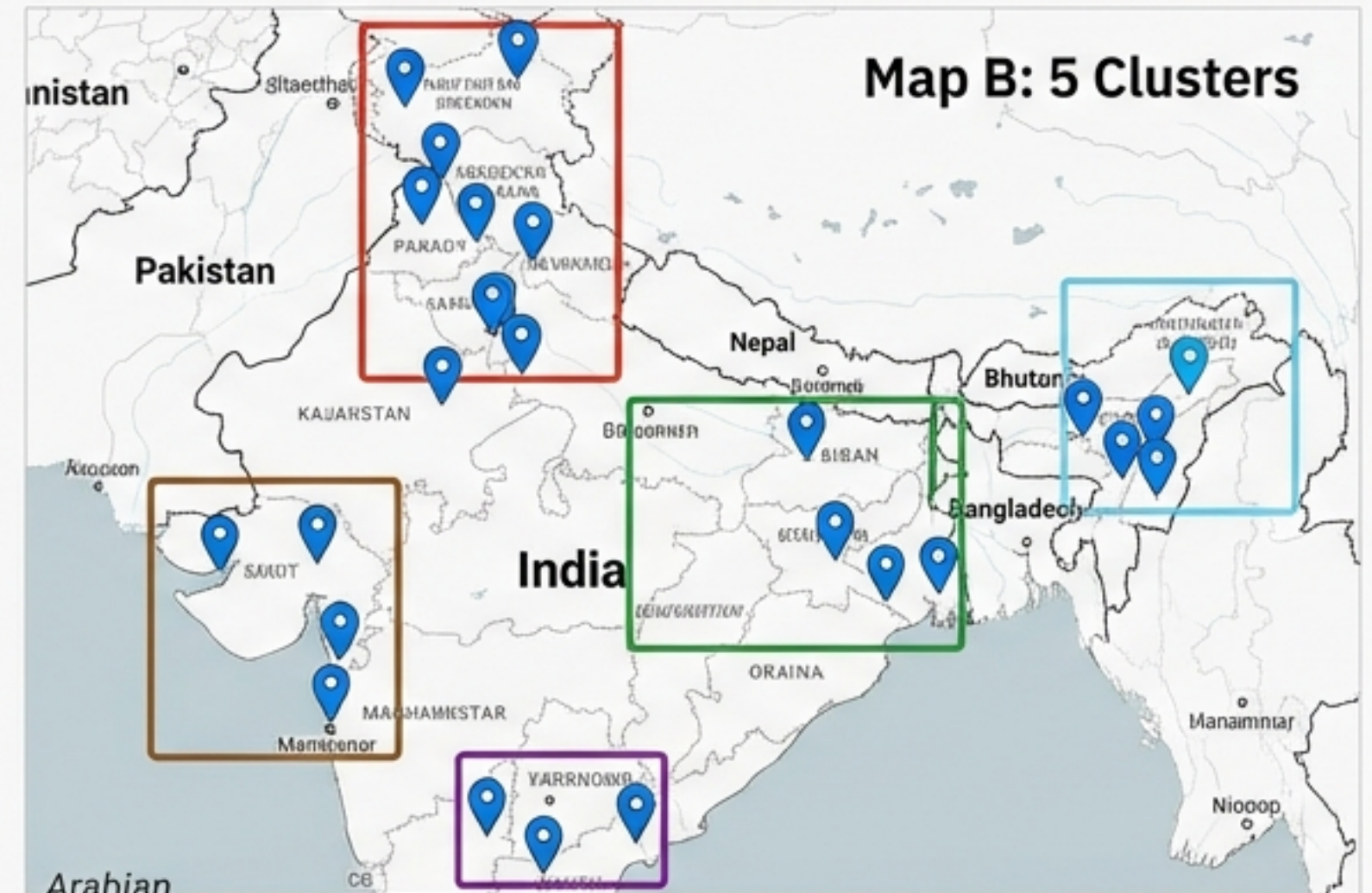
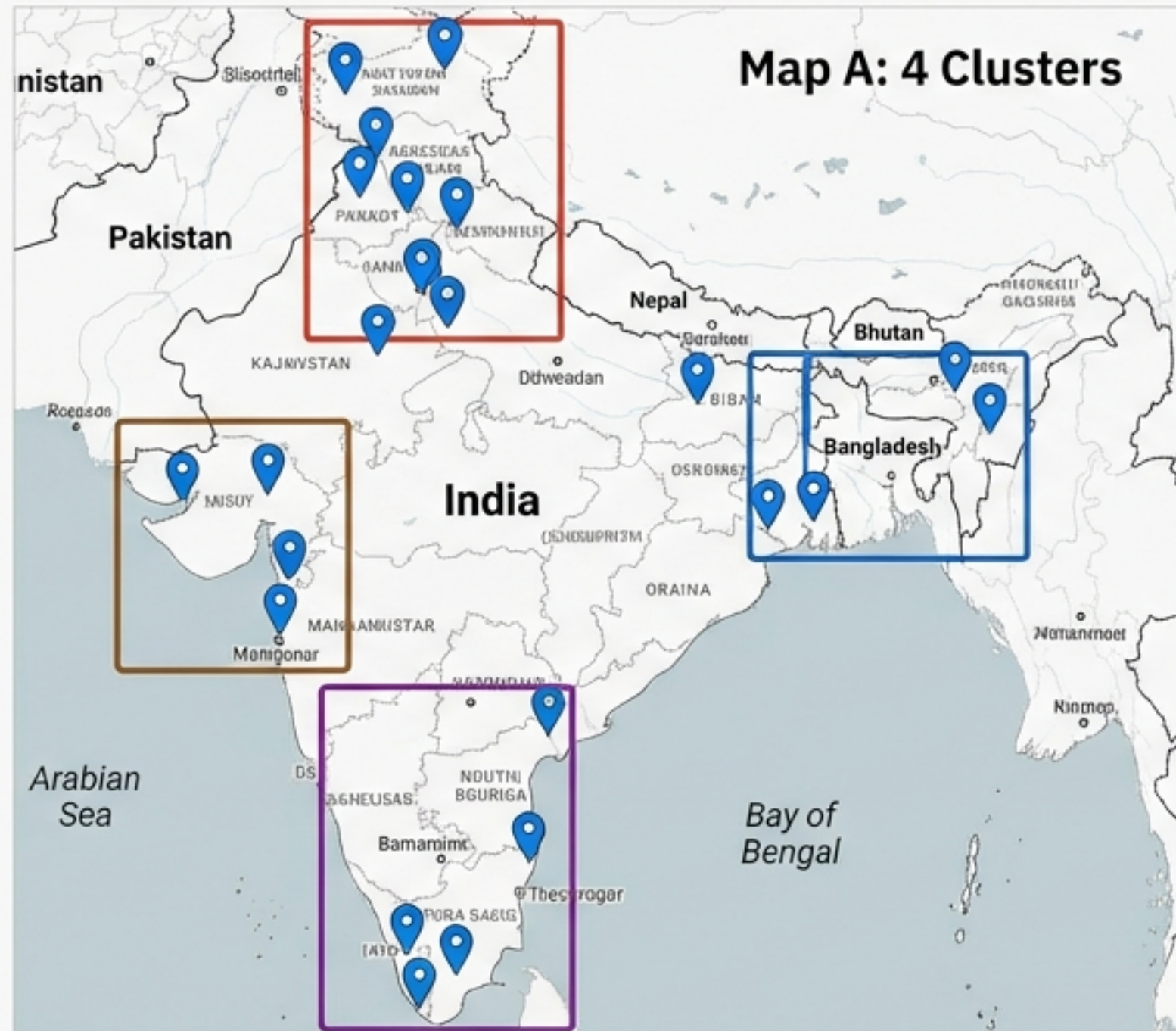


The Student View:
Students receive the dataset and the map visualisation.

The Task: 'You are seeing 101 customer locations. Identify the clusters.'

Pedagogical Insight:
Visualisation allows students to spot density patterns immediately—North, South, East, West, and Central hubs—turning a spreadsheet into a geography problem.

The Strategic Challenge: Locating the Warehouses



The Prompt:

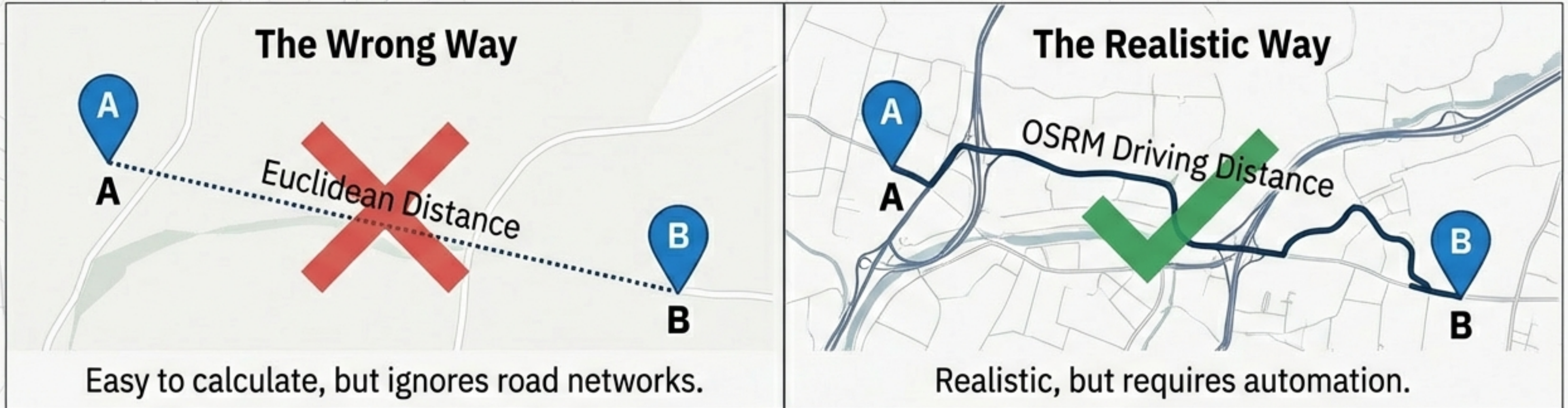
Recommend 3 to 5 warehouse locations to serve the network.

The Human Element:

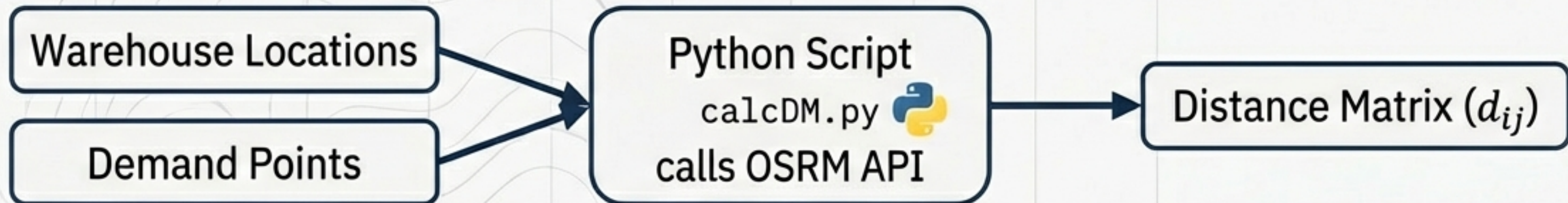
Students must move beyond math to consider:

- **Proximity:** Centroids of demand (Visual or K-means).
- **Infrastructure:** Access to highways/transport hubs.
- **Policy:** State tax structures and SEZs.

The Bridge: From Euclidean to Real-World Driving Distances

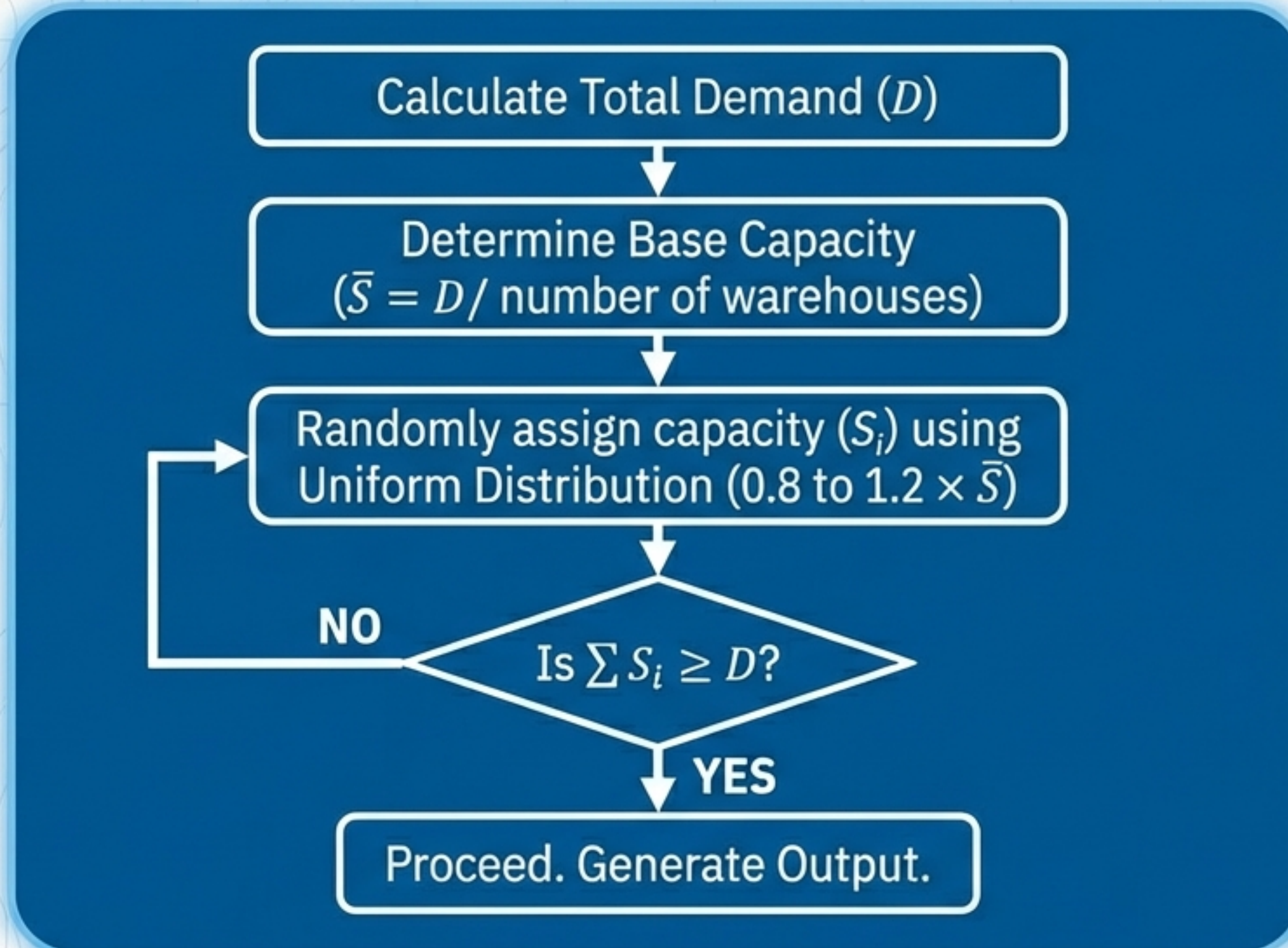


The Automation Solution (Instructor Python Script)



This automation allows the exercise to remain realistic without becoming tedious data entry work.

Ensuring Feasibility: The Capacity Algorithm



In a “No Shortage” model, Total Supply must be $\geq$ Total Demand.

This instructor-led step guarantees the student student models will be solvable.

Phase II: The Mathematical Formulation

Linear Programming Model

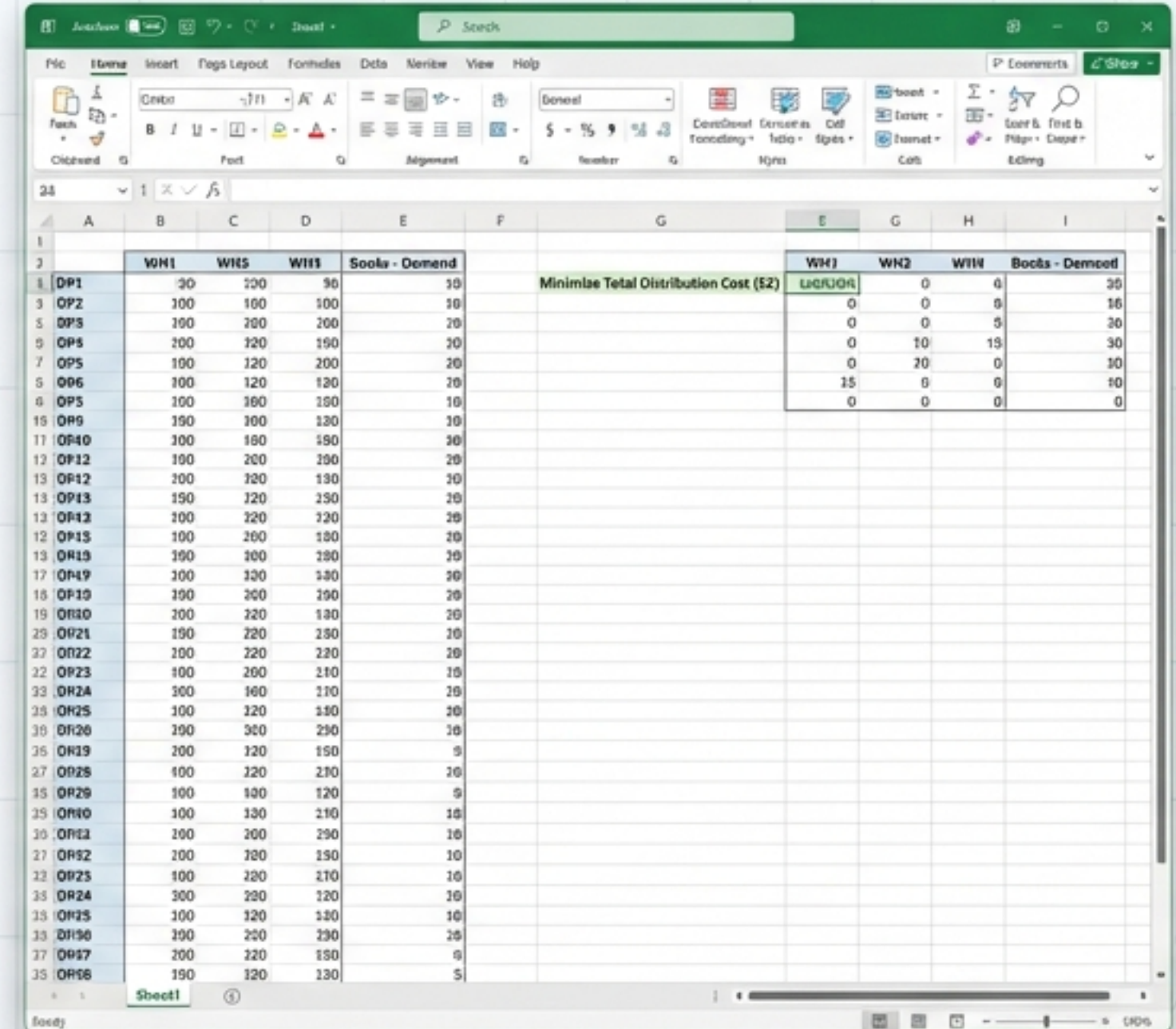
Objective: Minimize Total Distribution Cost (Z)

$$Z = \sum \sum c_{ij} x_{ij}$$

(where c_{ij} is cost based on driving distance)

Constraints:

1. **Supply:** $\sum x_{ij} \leq S_i$ (Outflow $\leq$ Capacity)
2. **Demand:** $\sum x_{ij} \geq D_j$ (Inflow $\geq$ Demand)
3. **Non-negativity:** $x_{ij} \geq 0$



	WH1	WH2	WH3	Stock	Demand
DP1	30	200	90	30	30
DP2	100	100	100	30	30
DP3	100	200	200	20	20
DP4	200	200	190	20	20
DP5	100	120	200	20	20
DP6	100	120	120	20	20
DP7	100	100	190	10	10
DP8	190	100	120	10	10
DP9	100	100	190	30	30
DP10	100	200	190	20	20
DP11	200	200	130	30	30
DP12	190	120	250	20	20
DP13	100	220	220	20	20
DP14	100	200	190	20	20
DP15	190	100	190	20	20
DP16	100	100	190	20	20
DP17	100	120	130	20	20
DP18	190	200	190	20	20
DP19	200	220	130	20	20
DP20	190	220	290	20	20
DP21	200	220	220	20	20
DP22	100	200	210	10	10
DP23	200	100	110	20	20
DP24	100	120	190	20	20
DP25	100	200	290	10	10
DP26	200	200	190	20	20
DP27	100	100	120	10	10
DP28	100	130	210	10	10
DP29	200	200	290	10	10
DP30	100	220	210	10	10
DP31	300	200	120	10	10
DP32	100	120	130	10	10
DP33	190	200	230	20	20
DP34	200	220	190	10	10
DP35	190	120	130	10	10

Scaling Up: Why We Need OpenSolver



Standard Excel Solver

Limit: 200 Decision Variables.

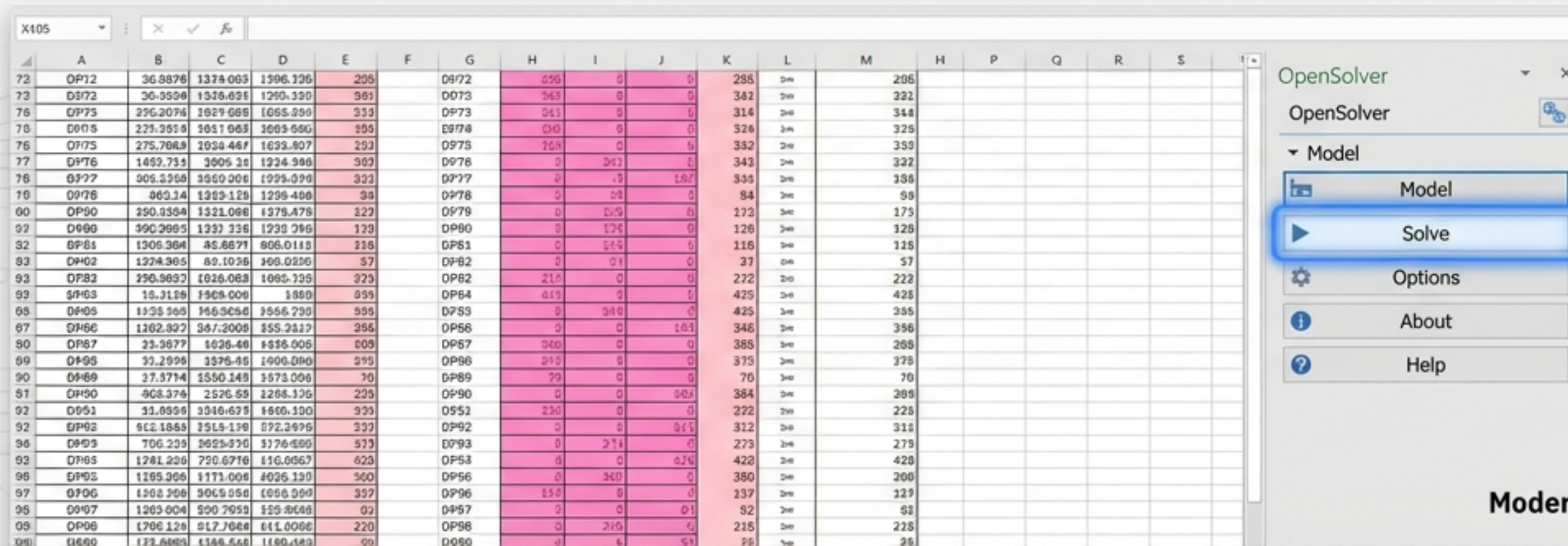
Insufficient for 100 demand points x 5 warehouses (500 variables).



OpenSolver

Limit: No artificial limits (Linear/Integer).

Why: Essential for solving real-world scale logistics problems.

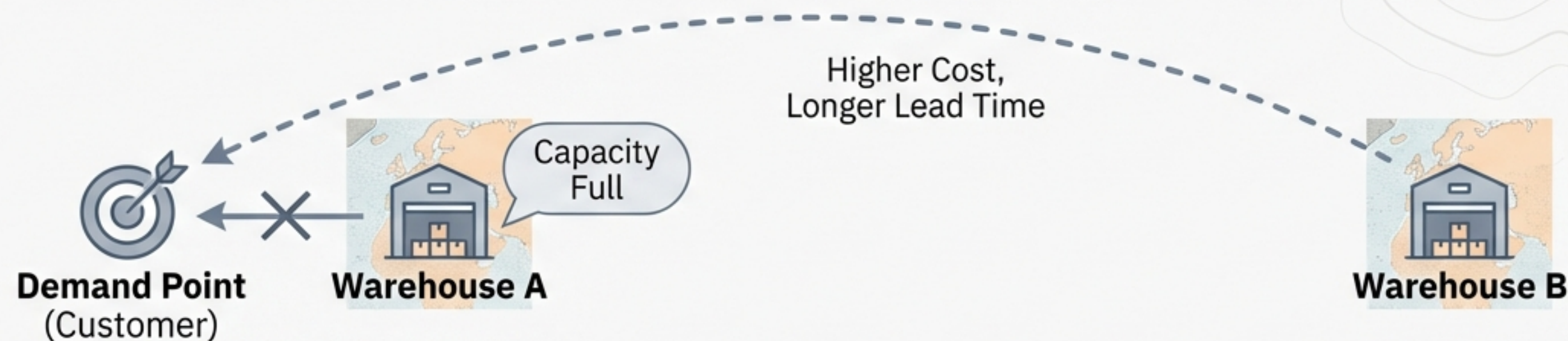


The screenshot shows an Excel spreadsheet with columns A through S and rows 72 through 99. The data is organized into a grid with alternating pink and white rows. The OpenSolver interface is visible on the right side of the screen, featuring a 'Model' tab, a 'Solve' button, and other options like 'Options', 'About', and 'Help'.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
72	OP12	36.8876	1374.063	1506.136	206		OP72	0.00	0	0	288	0.00	206						
73	OP72	36.8876	1374.063	1506.136	206		OP73	0.00	0	0	342	0.00	322						
74	OP73	276.2076	1829.686	1665.290	333		OP74	0.00	0	0	314	0.00	344						
75	OP74	223.3558	1651.065	1603.660	105		OP75	0.00	0	0	326	0.00	326						
76	OP75	275.7068	1934.467	1633.807	293		OP76	0.00	0	0	332	0.00	350						
77	OP76	1453.733	3605.26	1924.946	303		OP77	0.00	0	0	343	0.00	322						
78	OP77	905.3366	3950.006	1995.096	323		OP78	0.00	0	0	335	0.00	336						
79	OP78	863.14	1353.126	1296.486	34		OP79	0.00	0	0	84	0.00	98						
80	OP79	290.3364	1321.086	1378.476	223		OP80	0.00	0	0	173	0.00	173						
81	OP80	390.3665	1331.336	1233.086	133		OP81	0.00	0	0	126	0.00	126						
82	OP81	1306.364	45.6677	806.0115	218		OP82	0.00	0	0	116	0.00	116						
83	OP82	1324.365	89.1026	308.0256	57		OP83	0.00	0	0	37	0.00	57						
84	OP83	290.3632	1026.063	1065.135	323		OP84	0.00	0	0	272	0.00	222						
85	OP84	15.3186	1908.000	1880	899		OP85	0.00	0	0	425	0.00	425						
86	OP85	1335.366	1663.056	1666.230	999		OP86	0.00	0	0	425	0.00	355						
87	OP86	1102.392	367.2008	155.2112	396		OP87	0.00	0	0	346	0.00	396						
88	OP87	23.3677	1626.46	1536.006	609		OP88	0.00	0	0	385	0.00	208						
89	OP88	33.2995	1375.46	1400.000	399		OP89	0.00	0	0	373	0.00	378						
90	OP89	27.3714	1550.145	1673.006	70		OP90	0.00	0	0	70	0.00	70						
91	OP90	408.376	2626.69	1265.136	293		OP91	0.00	0	0	384	0.00	289						
92	OP91	31.8956	3316.625	1600.130	333		OP92	0.00	0	0	222	0.00	226						
93	OP92	612.1855	2515.136	872.2976	333		OP93	0.00	0	0	312	0.00	318						
94	OP93	706.233	3625.370	3776.666	573		OP94	0.00	0	0	273	0.00	279						
95	OP94	1241.226	720.6716	110.0067	623		OP95	0.00	0	0	428	0.00	428						
96	OP95	1165.366	1171.006	4036.130	360		OP96	0.00	0	0	360	0.00	200						
97	OP96	1165.366	3063.056	1056.390	337		OP97	0.00	0	0	237	0.00	123						
98	OP97	1263.604	300.7953	159.8668	02		OP98	0.00	0	0	92	0.00	63						
99	OP98	1706.126	517.7644	611.0066	220		OP99	0.00	0	0	218	0.00	218						
100	OP99	173.6669	1146.646	1180.189	09		OP100	0.00	0	0	26	0.00	26						

The "Aha!" Moment: Cost vs. Responsiveness

The Insight: Students realise the “closest” warehouse is not always the supplier.

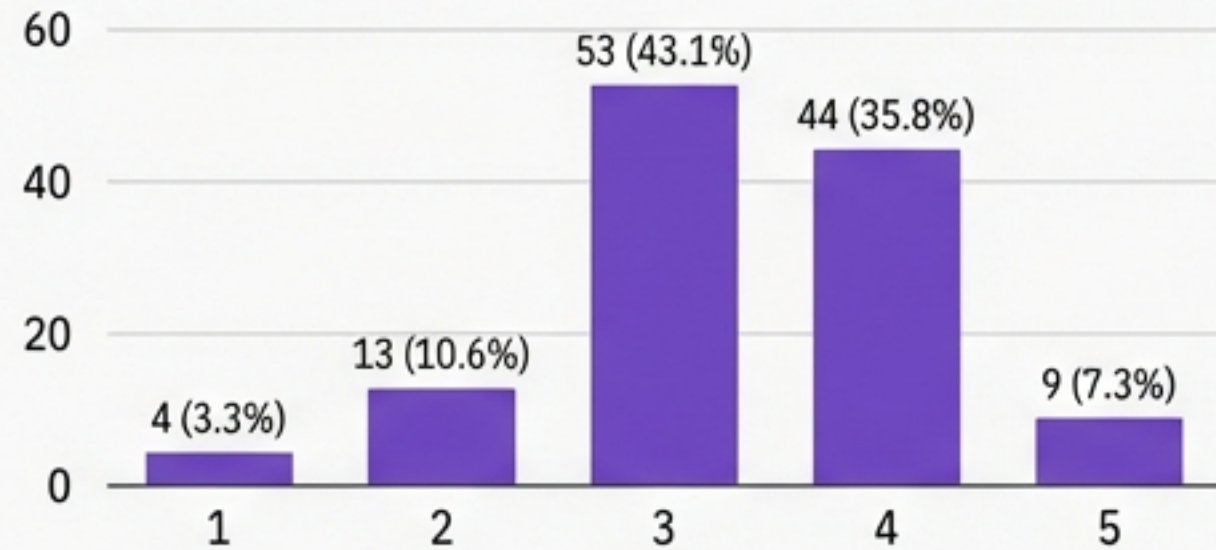


Summary Results: Cost Variance Based on Warehouses

Group	Objective Function Value (Total Cost)	# of Warehouses
1st Group 4	INR 14,889,335.00	3
2nd Group 8	INR 12,111,353.00	4
3rd Group 12	INR 12,737,260.00	4
4th Group 16	INR 12,256,156.00	4

Student Feedback & Pedagogical Wins

Prior Quantitative Exposure



Stream	
Engineering	66.7%
Commerce	9.8%
Economics	6.5%
Management	6.5%
Agriculture	4.8%
Science	4.1%
Biotechnology	1.6%
Total	100.0%

The assignment simulated real life problems... [and] helped in understanding perspectives of team members.

We could see the real locations... realising how important and effective OR methods can be.

Key Takeaways: Ownership of data, Integrated Strategic/Tactical decision making, and Visualisation as an analytical tool.

Implementation Guide for Instructors



Setup

- Run as a group assignment to foster collaboration.
- Use Python for the “heavy lifting” of distance calculations to maintain momentum.



Discussion Points

- Compare groups with 3 vs. 5 warehouses.
- Discuss the trade-off: Fixed Costs (Opening Facility) vs. Variable Costs (Transportation).



Future Extensions



- **Storage Compatibility:** Restrict storing food with chemicals.
- **Minimum Quantities:** Introduce binary variables for minimum flow constraints.

